Throughout this program I've learned and developed skills in different areas, including software development, problem solving, databases, and security. All the assignments I've done have given me experience in working with different programming languages, improving on applications, and finding solutions to any technical problems that I find. It also helped me confirm my decision on what I want to do professionally which is to get a career in app or web development where I can use all the skills I learned to make applications that are useful and easy to use.

I've learned that in order to be successful in developing software, communication and collaboration are an important part of the process. Working with others, understanding the project and its requirements, and communicating with stakeholders help make sure that everyone is working towards the same goal. Throughout this course I feel like I've learned how to do that, I've practiced documenting my code, have worked on how to explain my technical decisions, and I've learned to review and understand feedback so I can improve on my work. All of these experiences have taught me how important it is to make sure everyone involved understands what is being developed and why.

The way I understand software engineering, algorithms, and data structures is a lot better now and I've learned how to organize code to make applications easier to maintain and how to choose the right algorithms and data structures in order to make the performance better. While working with databases I learned how to store, retrieve and manage information and also have gained experience with testing and debugging in order to find errors and improve applications.

Security is another area I've learned a lot about and have gained some experience in. In one of my courses I evaluated a company's plan to use AI when it came to their hiring process. I found potential problems when it comes to bias, security, and transparency and then recommended how those things can be improved upon by doing additional testing, some training, and better communication with applicants. This helped me understand why developers need to consider privacy and vulnerabilities and how technology can affect the people that are using it.

For this course I chose three projects that allowed me to show my skills and what I've learned in different areas of computer science. My first one was a 3D scene where I made a reusable function to reduce a lot of extra unnecessary code and improved organization. The second one was a course planner where I added maps, graphs, cycle detection, and topological sorting to improve how the courses and perquisites were processed. My third project was for an animal shelter dashboard where I added pagination, search functionality, and better queries to improve how the information was taken and displayed.

All of these show my ability to work with different types of software and that I know how to improve their design, functionality, and performance.